

Tejas Mane

Ph.D. Candidate, Computer Science, University of California, Riverside

[Portfolio Website](#) | [GitHub](#) | [Google Scholar](#)

Education

University of California, Riverside Ph.D. in Computer Science, Advisor: Prof. Mohsen Lesani GPA: 3.49/4.0 Focus: Distributed Systems, Fault Tolerance, Consensus Protocols, Real-Time Scheduling	Aug 2021 – Dec 2026
University of Pennsylvania, Philadelphia M.S.E. in Scientific Computing GPA: 3.5/4.0	Aug 2018 – May 2020
BITS Pilani B.E. in Mechanical Engineering GPA: 8.48/10.0	Aug 2013 – May 2017

Research & Engineering Experience

Graduate Research Assistant — University of California, Riverside - Designed and implemented HAMAVA, a fault-tolerant geo-replicated database supporting heterogeneous clusters, arbitrary failures, and dynamic reconfiguration; improved throughput by up to 5× over HotStuff-based baselines and ~80% over BFT-SMaRt-based designs across 12 geo-distributed GCP clusters. - Implemented gossip-based metadata replication in Ray to recover from owner failures, distributing object-location and dependency state across workers so surviving nodes can reconstruct lost execution metadata and resume dependent computations without centralized coordination; optimized the protocol to limit overhead to about a 5% throughput drop. - Designed Shabdiz, an unauthenticated Byzantine consensus protocol for heterogeneous quorum systems; introduced conditional-ready messages for safe state pruning while preserving a 3-round common-case fast path. - Developing a vector-clock-based scheduler for ROS2 callback DAGs in cyber-physical and real-time systems, preserving causality across dependent callbacks while targeting avoidable scheduling delays beyond prior linear-chain scheduling models.	Aug 2022 – Present
Research Assistant — University of Pennsylvania - Built machine learning and computer vision pipelines for clinical hair-loss analysis, including hair-density estimation, segmentation, and web-based monitoring in collaboration with dermatology researchers. - Developed a single-camera 3D morphable-model fitting pipeline evaluated on 25 subjects; work was published in <i>Computer Vision and Image Understanding</i> and presented at the CVPR 2022 Workshop on Computer Vision for AR/VR.	Aug 2019 – Aug 2021
Computer Vision / Machine Learning Intern — Characterfacegen, Vidalign Inc. - Accelerated a 3D Morphable Model fitting pipeline from 1 FPS to 6 FPS using OpenMP parallelism, optical-flow stabilization, Python, and OpenCV. - Extracted lighting-normalized textures and pure-color maps for facial modeling and rendering pipelines.	May 2019 – Aug 2019
Junior Research Fellow — IIT Gandhinagar Conducted computational materials science simulations using molecular dynamics and density functional theory.	Aug 2017 – Jul 2018

Selected Publications

- Hamava: Fault-Tolerant Reconfigurable Geo-Replication on Heterogeneous Clusters*. IEEE ICDE 2025.
- Automating Hair Loss Labels for Universally Scoring Alopecia from Images*. JAMA Dermatology, 2023.
- Single-Camera 3D Head Fitting for Mixed Reality Clinical Applications*. CVIU 2022; presented at CVPR 2022 Workshop on Computer Vision for AR/VR.
- 8+ peer-reviewed publications; 100+ citations. Full list: [Google Scholar](#).

Technical Skills

Languages: C++, Python, Java, SQL

Distributed Systems: Consensus protocols, Byzantine fault tolerance, HotStuff, PBFT, Raft, Stellar/SCP, geo-replication, reconfiguration, gossip protocols, distributed futures, Ray

Machine Learning / Computer Vision: PyTorch, TensorFlow, scikit-learn, OpenCV, federated learning, Flower

Infrastructure: Linux, Docker, Kubernetes, Git, CMake, OpenMP, Slurm, Google Cloud Platform, AWS

Achievements & Awards

- Student Travel Grant, IEEE ICDE 2025.
- Licensed two machine learning and computer vision systems to a startup, May 2024.
- Research featured by Dermatology Times for alopecia scoring technology, March 2023.
- KVPY Fellowship recipient, top 500 nationally, India.